Korea Income and Welfare

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Education premiums, predictive benchmarking, and distributional heterogeneity in Korean welfare microdata

Abstract

This brief studies whether education still carries measurable income value once household size, age, region, gender, religion, and survey year are added to the analysis. The repository combines descriptive summaries, non-parametric tests, machine-learning benchmarking, semilog OLS with robust inference, and quantile regression. The central result is that education has a clearly positive average premium, but that premium is not constant across the income distribution. Machine learning improves pure prediction materially, while the econometric layer explains what those predictive gains mean.

1 Question

The project asks three linked questions:

1. Does education show a strong and statistically meaningful relationship with income in Korean welfare microdata?
2. How much does predictive performance improve once richer demographic and household information is added?
3. Is the education premium constant, or does it vary across lower-income and higher-income parts of the distribution?

The analysis is intentionally descriptive and predictive rather than causal. The goal is to measure signal, compare model behavior honestly, and explain the income premium in a way that is useful for analytics, labor-market research, and public-policy storytelling.

2 Data and empirical design

The raw dataset contains 92,857 survey records. After removing invalid income observations, restricting the age range to 18 through 90, and keeping rows with usable education information, the main analysis sample contains 89,935 observations. A standard interquartile-range trim on income yields a final modeling sample of 86,313 observations.

The workflow has five layers:

1. Descriptive EDA across education, region, and survey year.
2. Formal distribution tests using Kruskal-Wallis, Spearman rank correlation, and Mann-Whitney U.

3. Predictive benchmarking with education-only linear regression, multivariable linear regression, Elastic Net, Gradient Boosting, and a tuned Random Forest.

4. A semilog OLS earnings equation with centered age terms and HC3 robust standard errors.

5. Quantile regression at the 25th, 50th, and 75th percentiles to test heterogeneous education returns.

3 Headline results

Indicator	Value
Raw dataset size	92,857
Analysis sample size	89,935
Trimmed modeling sample size	86,313
Spearman education-income association	0.609
Average OLS education premium per step	12.47%
Best holdout model	Tuned random forest ($R^2 = 0.645$)
Best cross-validated model	Tuned random forest ($CV\ R^2 = 0.649$)
Top region by median income	Seoul (USD 2,883.64)

The non-parametric tests reject the idea that the education-income relationship is noise. Income distributions differ strongly across education groups, the education rank correlation is large and positive, and raw male and female income distributions remain statistically different before conditioning on controls.

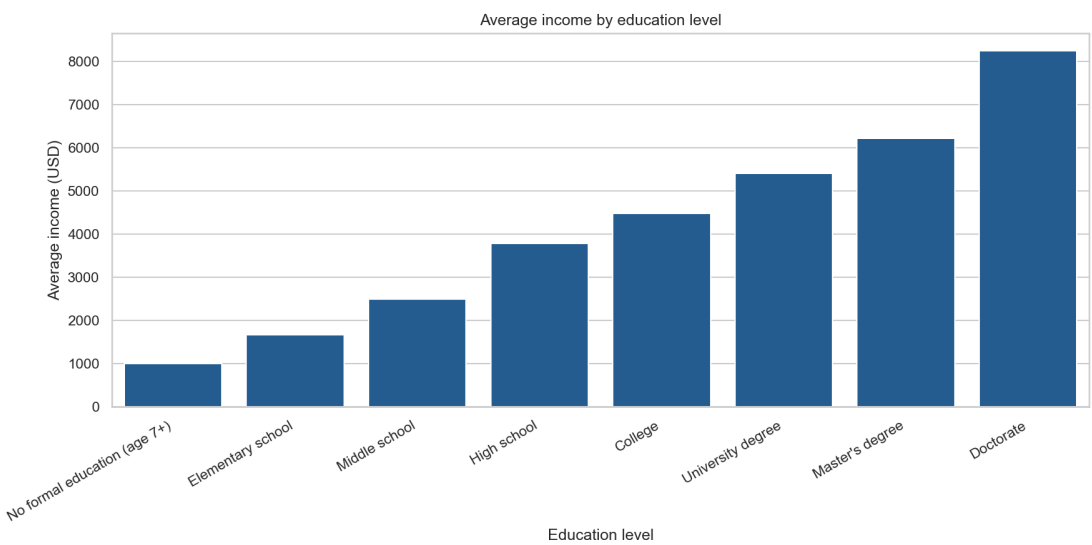

Figure 1: Average income rises monotonically with education level, from roughly USD 1,000 among the least educated group to more than USD 8,000 among doctorate holders.

Specification	Education premium	Fit metric
Semilog OLS	12.47%	—
Quantile regression, 25th percentile	11.46%	pseudo-$R^2 = 0.418$
Quantile regression, 50th percentile	13.84%	pseudo-$R^2 = 0.428$
Quantile regression, 75th percentile	14.17%	pseudo-$R^2 = 0.385$

This pattern matters because it shows that education is not acting like a uniform scalar reward. The return is positive everywhere, but it becomes larger in better-paid parts of the sample, which is consistent with labor-market sorting rather than a flat premium.

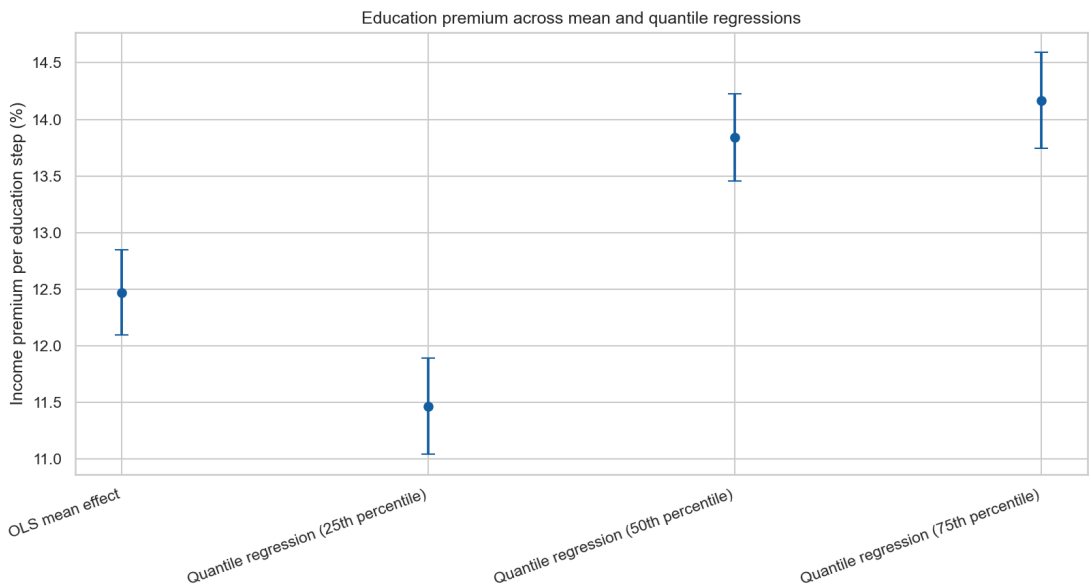

Figure 2: The education premium is positive across all specifications, but it increases toward the upper part of the income distribution.

4 Predictive benchmarking

Pure prediction improves materially once the feature set moves beyond education alone.

Model	Holdout R^2	Cross-validated R^2
Education-only linear regression	0.294	0.294
Multivariable linear regression	0.545	0.542
Elastic Net	0.545	0.542
Gradient Boosting	0.603	0.599
Tuned random forest	0.645	0.649

The tuned random forest is the strongest predictor, but the econometric layer remains essential because it converts a higher score into an interpretable claim. The feature-importance ranking

shows that household size, age, education, and survey year explain most of the incremental predictive power, while the OLS and quantile models explain how the education premium behaves.

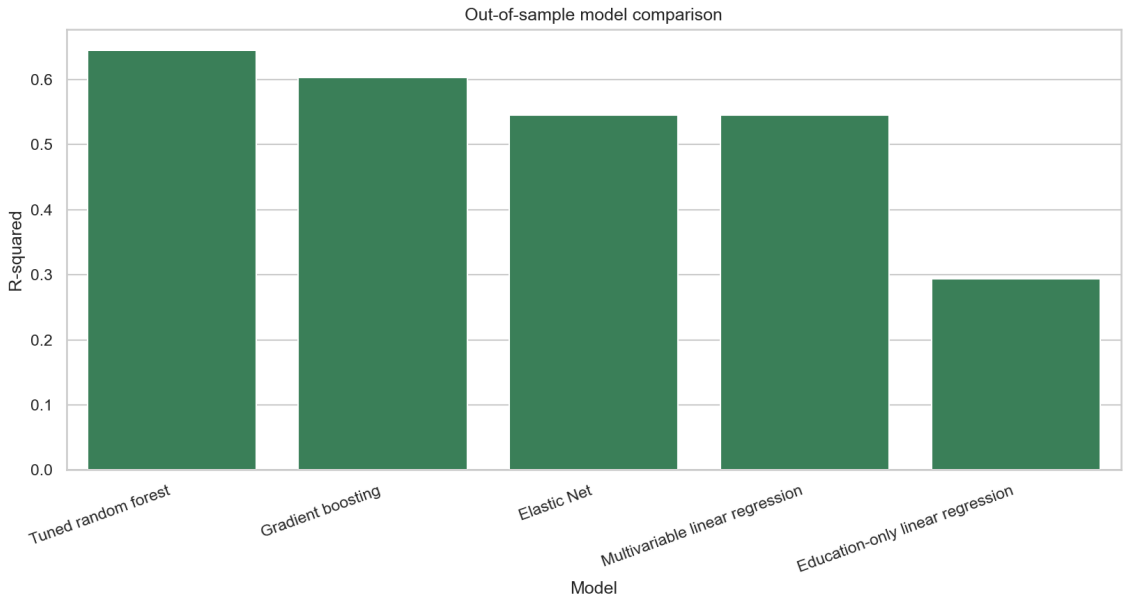

Figure 3: Adding household and demographic context materially improves predictive performance beyond an education-only regression.

5 Diagnostics and interpretation

The semilog specification is not used blindly. Heteroskedasticity is present, so the main OLS table is reported with robust errors. Centering the age polynomial keeps VIF values in a low range and stabilizes the nonlinear life-cycle profile. RESET rejects the idea that a single simple linear specification is fully sufficient, which is exactly why the repository pairs the OLS layer with machine learning and quantile regression instead of presenting a single coefficient table as the whole story.

Taken together, the repository supports a clear conclusion: education has a strong positive average association with income, but that return is heterogeneous across the distribution. Richer socioeconomic context improves prediction substantially, yet interpretation still depends on the econometric layer.

6 Repository map

The analysis is organized around four entry points:

- `income_welfare_modeling.ipynb` for the narrative walkthrough.
- `analysis_final.py` for regenerating figures and tables from the raw CSV.
- `outputs/figures` and `outputs/tables` for reusable portfolio assets.
- `docs/korea-income-welfare-brief.pdf` for the final technical summary.

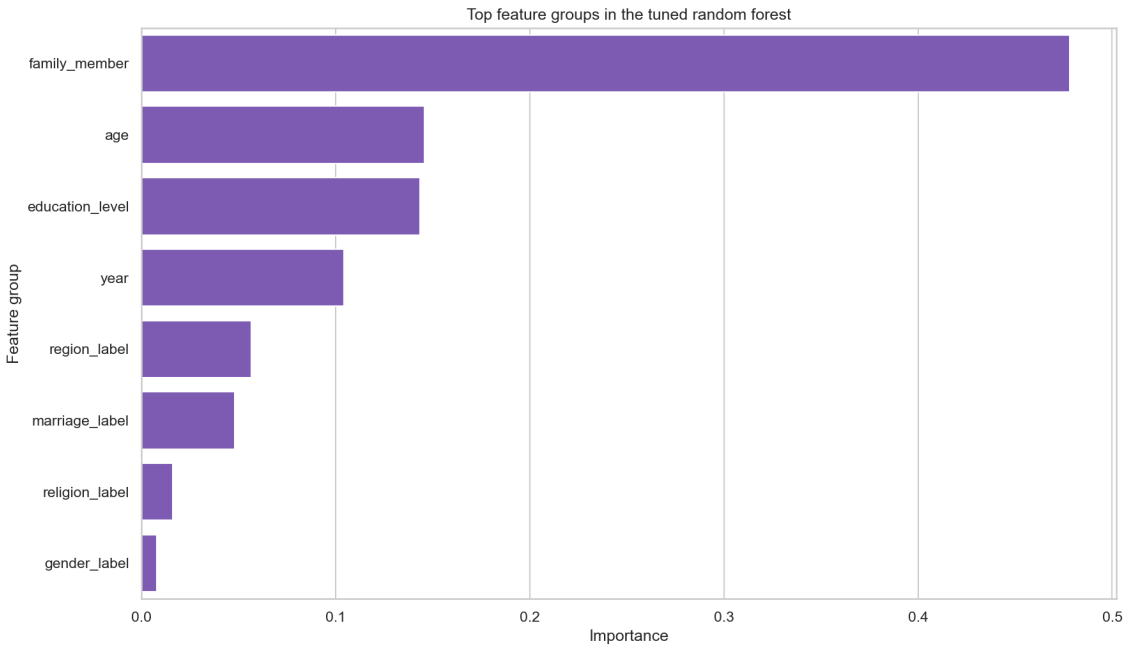

Figure 4: The tuned random forest relies most heavily on household size, age, education, and survey year, with gender contributing comparatively little incremental signal.